Why

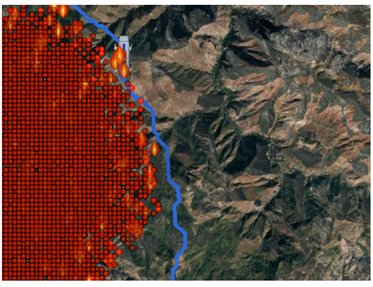
The edge of a wildfire is one of the most dangerous places for a firefighter.

Robot dogs (four-legged robots) can climb steep, rocky ground where trucks get stuck, and measure conditions on the ground under smoke and trees.

But a dog that steers by where the fire is now can walk into where it will be.

Plan on the predicted fire

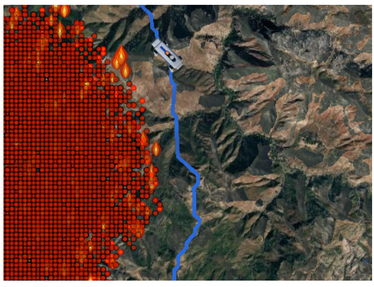
Planned on the fire now



In the flames

75 min in, the fire has reached the route.

Planned on the prediction



Still clear

Same moment: this route keeps 100 m from the predicted fire.

The dog plans to stay at least 100 m from where the fire is predicted to be, with 30 min in hand.

It scouts from the closest spot that keeps that distance and still has a way home. When its sensors show the real fire differs, the prediction reruns and the dog re-plans. If it is cut off, an emergency retreat may come closer.

Simulator screenshots, Big Sur: same fire, same start and target, 75 minutes after the dog sets out.

How we predict the fire

Fire history + conditions

246,000 NASA satellite fire detections from 30 wildfires; weather, terrain, land cover

→

Machine learning + fire simulator

XGBoost gives each spot's chance of catching fire; the simulator spreads it with the wind

→

When each spot burns

Every 50 m spot, in 15 min steps

Small and fast, so the dog can re-plan on the spot.

31 KB

The whole model. Smaller than one photo on this poster.

0.1 s

One model pass over all 360,000 spots on the map (desktop CPU).

No GPU

No internet needed once loaded. Built for small edge computers; Raspberry Pi tests are next.

Model trained to predict whether a fire keeps spreading on 24 fires, tested on 6 others (AUC 0.75). The 50 m, 15 min spread is a simulation, not checked against real fires.

Close enough to help, safe enough to return

430 simulated missions per dog

	Reached a spot to scout	Caught by the fire
Reacts, keeps 500 m away	0%	0%
Reacts, keeps 200 m away	27%	0.2%
Predicts, no margin	100%	46%
Predicts, 100 m + 30 min + sensing	91%	0.5%

Reacting stays safe by staying back